

Course Code	: 2304CS304	Date	: 22-04-2026
Course Name	: Software Engineering	Duration	: 150 Minutes
		Total Marks	: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

- Q.1 (A)** Define Software Engineering. Explain characteristics of software. **4**
- (B)** Explain Software Development Life Cycle (SDLC) phases. **3**

OR

Explain Software Engineering as a layered technology.

- (C)** Compare Waterfall Model and Spiral Model with advantages and disadvantages. **7**

OR

Explain Agile model and Scrum framework in detail.

- Q.2 (A)** Define Functional and Non-Functional Requirements with examples. **4**
- (B)** What is SRS? List characteristics of a good SRS. **3**

OR

Explain Requirement Engineering process.

- (C)** Explain Use Case Diagram and Sequence Diagram with examples. **7**

OR

Explain Data Flow Diagram (DFD) and Class Diagram in detail.

- Q.3 (A)** Define Cohesion and Coupling with its Types. **4**
- (B)** Explain Software Design Principles. **3**

OR

Compare Function-Oriented and Object-Oriented Design.

- (C)** Explain Architecture Design pattern in details. **7**

OR

Discuss Golden rules of User Interface Design.

- | | | | |
|------------|------------|--|----------|
| Q.4 | (A) | Explain Coding Standards and Guidelines. | 4 |
| | (B) | Explain White Box Testing and Black Box Testing. | 3 |

OR

Explain Integration Testing.

- (c)** Describe Software Testing Strategies in detail. **7**

OR

Explain Validation Testing and System Testing with examples.

- | | | | |
|------------|------------|--|----------|
| Q.5 | (A) | Define Software Project Management and its responsibilities. | 4 |
| | (B) | Explain Line of Code (LOC) estimation. | 3 |

OR

Explain Function Point Analysis.

- (c)** Explain COCOMO model in details. **7**

OR

Discuss Software Project Planning, Scheduling and Tracking.
